MC LOCAL POWER — IS-2 HARDWARE PROPOSAL / NOT MANUFACTURING READY

CORE_1V2: 1.2 V / 0.8 A reservation; 1 MHz forced PWM

IO_3V3: 3.3 V / 0.15 A reservation; 2.5 MHz forced PWM

CORE_5V
C1
10uF 25V X7R 10%
LOGIC_GND

U1
TPS62901RP/JR
CORE_5V
CORE_1V2_EN
CORE_1V2_MODE
CORE_1V2_SS
VIN
EN
MODE/S-CONF
SS/TR
SW
PG
VOS
FB/VSET
LOGIC_GND
CORE_1V2_SW
CORE_1V2_PG_OD
CORE_1V2

L1
CORE_1V2_SW
2.2uH
CORE_1V2

CORE_1V2
C2
22uF 10V X7S 20%
LOGIC_GND

CORE_1V2_MODE
R1
76.8k 1%
LOGIC_GND

CORE_1V2_EN
R2
10k 1%
LOGIC_GND

CORE_1V2_SS
C3
6.8nF 50V C0G 5%
LOGIC_GND

CORE_1V2
R3
100k 1%
LOGIC_GND

CORE_5V
C4
10uF 25V X7R 10%
LOGIC_GND

U2
TPS62901RP/JR
CORE_5V
IO_3V3_EN
IO_3V3_MODE
IO_3V3_SS
VIN
EN
MODE/S-CONF
SS/TR
SW
PG
VOS
FB/VSET
LOGIC_GND
IO_3V3_SW
IO_3V3_PG_OD
IO_3V3
IO_3V3_VSET

L2
IO_3V3_SW
1uH
IO_3V3

IO_3V3
C5
22uF 10V X7S 20%
LOGIC_GND

IO_3V3_MODE
R5
32.4k 1%
LOGIC_GND

IO_3V3_EN
R6
10k 1%
LOGIC_GND

IO_3V3_SS
C6
6.8nF 50V C0G 5%
LOGIC_GND

IO_3V3
R7
100k 1%
LOGIC_GND

IO_3V3_VSET
R8
301k 1%
LOGIC_GND

CORE_5V
R21
97.6k 0.01% 5ppm
CORE_5V_UV_SENSE

CORE_5V
R29
92.1k 0.01% 5ppm
CORE_5V_OV_SENSE

CORE_5V
C7
100nF 25V X7R 10%
LOGIC_GND

U3
LTC2962HUD#TRPBF
CORE_5V
CORE_5V_UV_SENSE
CORE_5V_OV_SENSE
LOGIC_GND
VCC
V1
V2
V3
V4
MR
RDIS
DVCC
RST
REF
PG1
PG2
PG3
PG4
BT
GND
MC_SEQ_5V_OK
MC_SEQ_REF1
MC_SEQ_REF1
MC_SEQ_REF1
LOGIC_GND
MC_SEQ_REF1
MC_SEQ_REF1
CORE_5V

CORE_1V2
R41
1.07k 0.01% 5ppm
MC_SEQ_1V2_1

CORE_1V2
R49
1.15k 0.01% 5ppm
MC_SEQ_1V2_1

CORE_5V
C8
100nF 25V X7R 10%
LOGIC_GND

MC_SEQ_1V2_1
R42
261 0.01% 5ppm
CORE_1V2_UV_SENSE

MC_SEQ_1V2_1
R50
316 0.01% 5ppm
MC_SEQ_1V2_2

MC_SEQ_1V2_2
R51
4.02 1% 100ppm
CORE_1V2_OV_SENSE

U4
LTC2962HUD#TRPBF
CORE_5V
CORE_1V2_UV_SENSE
CORE_1V2_OV_SENSE
CORE_5V
CORE_5V
MC_SEQ_5V_OK
CORE_5V
LOGIC_GND
VCC
V1
V2
V3
V4
MR
RDIS
DVCC
RST
REF
PG1
PG2
PG3
PG4
BT
GND
IO_3V3_EN
MC_SEQ_REF2
MC_SEQ_REF2
LOGIC_GND
MC_SEQ_REF2
MC_SEQ_REF2
CORE_5V

IO_3V3
R61
54.9k 0.01% 5ppm
IO_3V3_UV_SENSE

IO_3V3
R69
57.6k 0.01% 5ppm
IO_3V3_OV_SENSE

CORE_5V
C9
100nF 25V X7R 10%
LOGIC_GND

U5
LTC2962HUD#TRPBF
CORE_5V
IO_3V3_UV_SENSE
IO_3V3_OV_SENSE
CORE_5V
CORE_5V
IO_3V3_EN
CORE_5V
LOGIC_GND
VCC
V1
V2
V3
V4
MR
RDIS
DVCC
RST
REF
PG1
PG2
PG3
PG4
BT
GND
MC_RAILS_OK_5V
MC_SEQ_REF3
MC_SEQ_REF3
LOGIC_GND
MC_SEQ_REF3
MC_SEQ_REF3
CORE_5V

CORE_5V_UV_SENSE
R27
10k 0.01% 5ppm
LOGIC_GND

CORE_5V_OV_SENSE
R35
10k 0.01% 5ppm
LOGIC_GND

CORE_1V2_UV_SENSE
R47
1k 0.01% 5ppm
LOGIC_GND

CORE_1V2_OV_SENSE
R55
1k 0.01% 5ppm
LOGIC_GND

IO_3V3_UV_SENSE
R67
10k 0.01% 5ppm
LOGIC_GND

IO_3V3_OV_SENSE
R75
10k 0.01% 5ppm
LOGIC_GND

CORE_5V
R101
3.3k 1%
MC_SEQ_5V_OK

MC_SEQ_5V_OK
R102
3.3k 1%
CORE_1V2_EN

CORE_1V2_EN
R103
10k 1%
LOGIC_GND

CORE_5V
R104
6.8k 1%
IO_3V3_EN

IO_3V3_EN
R105
10k 1%
LOGIC_GND

CORE_5V
R106
47k 1%
MC_RAILS_OK_5V

MC_RAILS_OK_5V
R107
470k 1%
LOGIC_GND

IO_3V3
MC_RAILS_OK_5V
U6
SN74LVC1G17DBVR
FPGA_CRESET_N
VCC
GND

FPGA_CRESET_N
R108
4.7k 1%
LOGIC_GND

IO_3V3
C10
100nF 25V X7R 10%
LOGIC_GND

B1
MC POWER REVIEW BOUNDARY
CORE_5V
LOGIC_GND
CORE_1V2
IO_3V3
FPGA_CRESET_N
MC_RAILS_OK_5V
CORE_1V2_PG_OD
IO_3V3_PG_OD
LOGIC_GND

5 V window releases core; core window releases IO; IO window releases FPGA reset.
Each supervisor releases 160–240 ms. Raw MC_RAILS_OK_5V is not a 3.3 V signal.
FPGA_CRESET_N is the buffered rails-good term; combine other reset requests upstream.
Regulator PG pins are diagnostic only; no PG-to-EN cascade. Physical connector is external.

Actual rail limits retained: 5 V 4.85–5.15; T20 core 1.15–1.25; IO 3.20–3.40 V.
Core thresholds include initial/temp plus 400ppm drift; historical tight ledger is distinct.
IO uses 2.5 MHz because existing FPGA decoupling exceeds the 1 MHz capacitance limit.
Proposed MC sum: 0.50 A steady / 0.80 A downstream startup input charging separate.
Arbitrary partial power, assembled stress, brownout slew, shutdown and timing unproved.

PWR_FLAG
CORE_5V
PWR_FLAG
LOGIC_GND
PWR_FLAG
CORE_1V2
PWR_FLAG
IO_3V3

Sheet: /
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